AUTOMATIC DETECTION OF LANDSLIDES BASED ON MACHINE LEARNING FRAMEWORK

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ABSTRACT

Advancement in the satellite imaging techniques has produced wider applications in disaster reduction and management. Considering the crucial role of rapid detection of landslides in disaster management, this research paper exploits the capability of sentinel-1A imagery in landslide detection. In image processing and feature detection techniques, machine learning based classification methods have gained popularity by proving its potential in mapping of susceptible landslides. Therefore, the primary objective of this research paper is automatic identification of landslides by using machine learning based classifier Random Forest (RF). To accomplish the objective, two Sentinel-1A GRD images, pre and post-event, covering Srinagar-Rudraprayag region, India has been processed with DEM derived products slope, aspect and elevation for landslide detection along the Alaknanda River and National Highway-58. For validation of the results, Google Earth Historic Imagery has been utilized. From the observation of results, RF method confirms its potential by achieving 95% in accurate detection of landslides.

Index Terms— Landslide, Machine Learning, Random forest, Natural Hazards

1. INTRODUCTION

Surface instability due to haphazard construction in hilly regions is a serious concern especially for lives of people, economy and ecosystem [1]. Although measures of safety and landslide prevention are carefully taken, but monitoring of old surface instabilities such as landslides and detection of newly developed landslides is necessary to reduce the loss and threat. The advancement in techniques and algorithms has fortunate us for surface monitoring with higher precision and wider coverage in comparison to traditional methods. Currently, we have a list of techniques based on the application such as GNSS for crustal movements with millimetre-level accuracy, laser scanners for surface deformation studies with high density points at millimetre precision and space and ground based synthetic aperture radar (SAR) for study of volcanic, earthquake and landslide activities with large coverage (swath) and higher accuracy [2-6].

Landslide susceptibility maps (LSM) are used to highlight the landslide prone regions in order to reduce risk associated with hazards. The accuracy of LSM depends on the factors included in preparation such as selection of appropriate thematic layers, scale and modelling technique (Moradi and Razaei, 2014). Applied modelling techniques include qualitative and quantitative, for example, fuzzy, analytic hierarchy process (AHP), fuzzy AHP, logistic regression, support vector machine (SVM) and random forest (RF) and neural networks [7-13]. Considering the complexity of landslide identification for wide regions, the objective of the study is to automate landslide classification process by machine learning classifier RF. The proposed method uses amplitude information of Sentinel-1A GRD images and DEM derived products slope, aspect and elevation as input features. The root cause of preferring SAR imagery is due to its capabilities of all time imaging and weather independency.

2. STUDY AREA

Himalayan arc contains a high percentage of landslide occurrences in Asia and a big portion of this increased in landslide is caused by haphazard road constructions and preventive measures [14]. Rudraprayag is one of most affected Indian states in Himalayan belt, site along the Alaknanda river and National Highway-58 in Srinagar-Rudraprayag region of Uttarakhand state are selected for the automatic classification of landslides shown in Figure 1 [15-16]. For the landslide classification, two Sentinel-1A GRD images have been utilized and the image acquisition dates are selected on the basis of increased landslide activity in the study region (Table 1).

3. METHODOLOGY

In the visual inspection of landslides from optical or SAR imageries, amplitude (intensity) change is an important source of information. In comparison of pre and post-event satellite imageries, a spike in amplitude change could be easily observed for the pixels of new landslides. For the automatic classification of landslides, the pixels of GRD images have been analysed. Therefore, two Sentinel-1A GRD images of pre-monsoon and post-monsoon have been acquired to detect the surface changes. Initially, the satellite acquisitions are pre-processed in Sentinel Application Platform (SNAP). Pre-processing includes orbital correction by precise position and velocity data of satellite, radiometric calibration and coregistration of both the scenes. Further, a subset of the coregistered scenes is created which covers the study area only. Using band Math operator, difference of amplitude is calculated and pixels of urban and water bodies are filtered. Finally, training and test data are created with amplitude difference band and landslide triggering factors such as slope, aspect and elevation. For automatic identification of landslides the training and test data is fed to machine learning classifier RF. The followed methodology is shown in Figure 2.

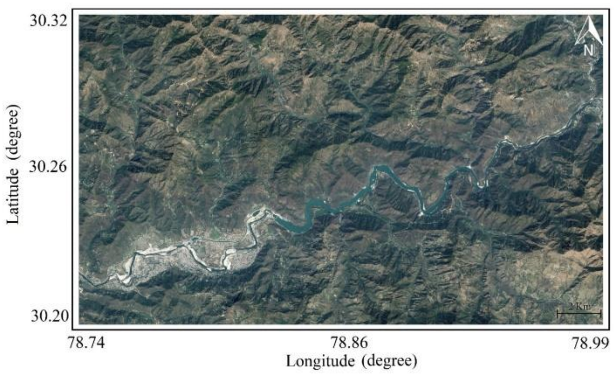

Figure 1: Study area: Srinagar-Rudraprayag region

Satellite image parameters	
Perpendicular base line (meter)	64.8
Temporal baseline (days)	192
Swath width (km)	250
Image acquisition dates	7-June-2018 & 16-Dec-2018

Table 1: Details of Sentinel-1A GRD images

3.1. Random Forest (RF)

RF is an ensemble of tree-structured classifiers which was introduced by [17]. The RF classifier utilizes numerous decision trees for class prediction. These decision trees are created based on the provided training dataset and each decision tree predicts an output. Each output is weighted by votes of class and widely held vote decides the final classification. The less sensitivity to over-fitting issues caused by increasing number of trees makes it favourable for classification of remote sensing images [13]. For our case, it could be specifically said as pixel classification. The classification problem is executed in two steps, training and prediction. Initially, a random sample of training pixels is selected to train each tree classifier and a split function is determined for all non-leaf nodes. The split parameters are selected in such way that it maximizes the gain function of classification [18], and finally statistics of the sampled training data is stored for each leaf node. For prediction, the probability of each pixel to belong to landslide or non-landslide class is computed for each tree and finally being assisted by votes of class, the pixel is classified as landslide or non-landslide point.

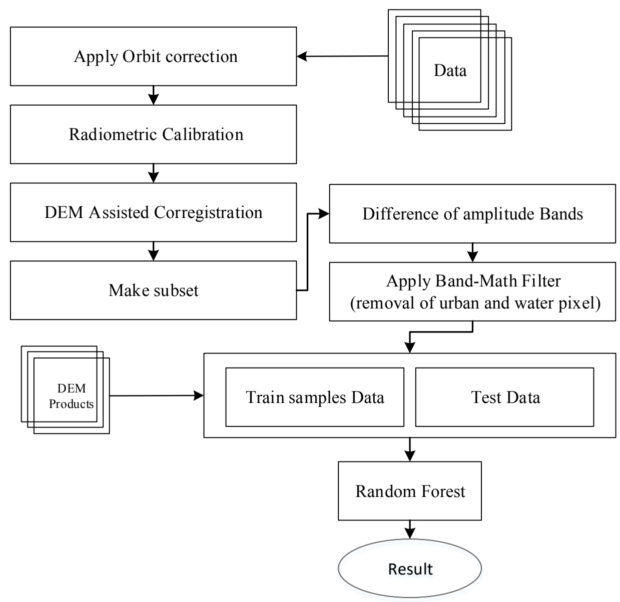

Figure 2: Flowchart for landslide identification using ML classification techniques

4. RESULTS AND DISCUSSION

Two GRD images, 7th June, 2018 and 16th Dec 2018, have been used for landslide identification using machine learning based classification technique. Using Band Math tool (Sentinels Application Platform) amplitude difference of each pixel is calculated. Histogram of difference of amplitude is shown in Figure 3. Based on the statistics of the amplitude differences of all the pixels and after an analysis of difference in amplitudes for the pixels available on the landslide, 105000 training samples are selected for training of classifiers, in which 100000 samples belong to non-landslide pixels and 5000 pixels are from landslides. For landslide identification and accuracy assessment of the classifiers, a total of 2300000 pixels are selected as test data. Random forest provides 528 points in the study area are identified as landslide points (Figure 4). Results are validated from the Google Earth's *Historic Imagery* tool (23 June, 2018 and 19 December, 2018) and shown in Figure 5 and 6.

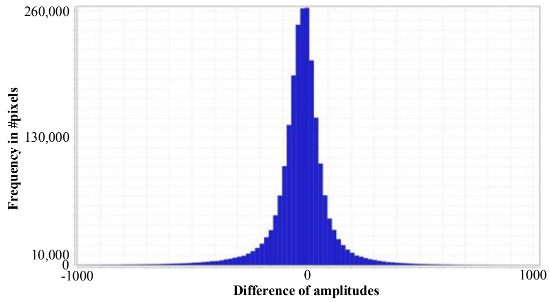

Figure 3: Histogram of difference of amplitude

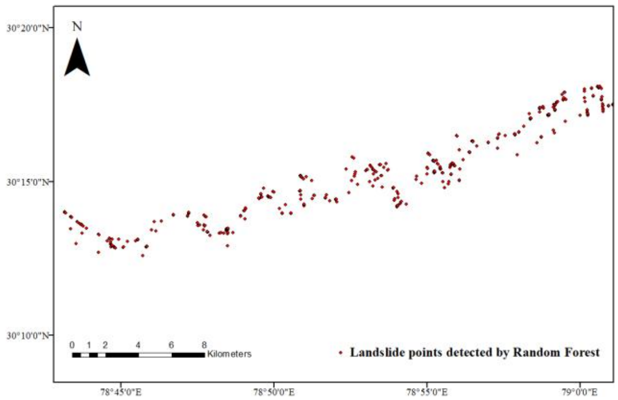

Figure 4: Pixels classified by Random Forest as landslide points

Figure 5: Validation of classification results in Google Earth using its *Historic Imagery* Tool.

Figure 6: Validation of classification results in Google Earth using its *Historic Imagery* Tool

5. CONCLUSIONS

This research work presents automatic classification of landslides using machine learning approach: Random Forest. After being trained by 105000 pixels (including landslide and non-landslide both), RF classifier has produced the highest classification result showing least sensitivity to training sample size. Results have been validated from optical image in Google Earth of same duration. Although the objective is landslide identification along the river, it is noticed that the algorithm has classified some pixels as landslide pixels which represents enough change in amplitude due to change in vegetation. In future, more landslide triggering features could also be included for automatic and more accurate landslide classification.

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